Project Title

Multi-Class Customer Category Prediction Using Multinomial Logistic Regression

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Machine learning (25SC2107E)

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ABSTRACT

Multi-Class Customer Category Prediction Using Multinomial Logistic Regression is a machine learning-based system designed to classify customers into different predefined categories based on their demographic, purchasing, and behavioural characteristics. In today's competitive business environment, understanding customer behaviour is important for improving marketing strategies, customer satisfaction, and business decision-making. The proposed system uses features such as age, income, purchase frequency, spending patterns, product preferences, and customer engagement to identify the category of a customer. Multinomial Logistic Regression is selected because it is suitable for classification problems involving more than two categories. The customer dataset is first pre-processed by handling missing values, encoding categorical attributes, and scaling numerical features when necessary. The processed data is then divided into training and testing datasets, where the training data is used to build the prediction model.

The performance of the model is evaluated using metrics such as accuracy, precision, recall, F1-score, and a confusion matrix to determine how effectively it classifies customers into their respective categories. The system can help businesses identify different types of customers and develop personalized marketing campaigns, product recommendations, offers, and customer retention strategies. By automating the customer classification process, the proposed approach reduces manual analysis and provides useful insights from large customer datasets. Overall, this project demonstrates how Multinomial Logistic Regression can be effectively applied to multi-class customer prediction and support data-driven business decisions.

Signature of the Guide